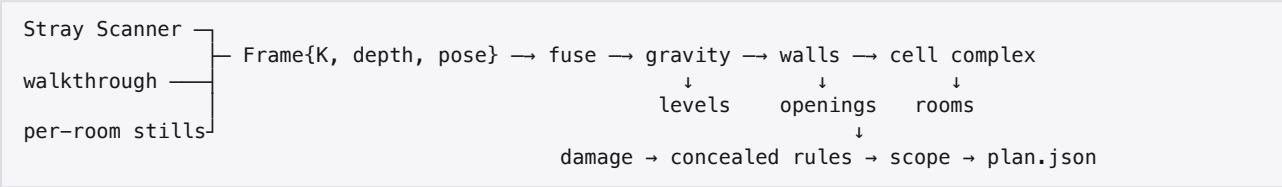


Cozmo — technical report

Handheld iPhone capture to a dimensioned floor plan, damage map and repair scope, at three input tiers. At most six pages, as specified; the page cap is why this omits things that went well and spends its space on the decisions and the numbers.

1. Architecture

One sentence governs the whole design: **a tier's job is to produce frames carrying intrinsics, metric depth and a pose. Everything after that is the same code.**



LiDAR is handed all three fields. Photo and video manufacture them — monocular depth, gravity from the floor plane, scale from camera height, pose from registration — and then call the identical `build_lidar_plan`. Three parallel reconstruction stacks would drift apart and only one would stay correct, and the tier comparison would then be measuring implementation differences rather than sensor differences.

The floor plan is a cell complex, not a raster. The wall lines partition the floor; each face is labelled interior or exterior from direct evidence (observed floor, carved free space, camera track). A flood fill was tried first and leaked through a glass balcony door, reporting 195 m² for a 44 m² flat. A face of an arrangement is bounded by lines on all sides, so a labelling mistake cannot propagate, and corners arrive as exact line intersections rather than staircases of pixels — which is what lets a room polygon inherit the millimetre-level offset uncertainty of the plane fits beneath it.

Walls come from a Hough accumulator over (normal azimuth, signed offset), each point voting once into the bin its own measured normal selects. Sharper than a classical line Hough, where every point smears a sinusoid across a cluttered accumulator. Peaks are accepted on observed surface area, not on a fraction of total vote weight: a forty-wall apartment divides its weight forty ways, so the strongest genuine peak carries 0.28% of the total and any fraction-of-total threshold admits everything or nothing.

Two wall representations are maintained deliberately. Tight segments, which stop at every gap, answer "is there material along this boundary" for the cell complex. Bridged runs answer "where are the holes" for opening detection. No single gap tolerance is both narrower than a doorway and wider than one, so trying to serve both from one representation means every doorway splits its wall and no segment then spans the door to notice it.

2. Tiers and the device matrix

Device	Photo	Video	LiDAR
iPhone 15/16/17 Pro, Pro Max	yes	yes	yes
iPhone 15/16/17 non-Pro	yes	yes	no sensor
iPhone 14 and older	out of contract	out of contract	out of contract

LiDAR. Stray Scanner export: ARKit poses, 256×192 depth in millimetres, per-frame intrinsics, ARKit's own confidence channel. Depth is weighted by inverse variance from a noise model of confidence class and range, never thresholded, so a low-confidence return still contributes in proportion to what it is worth. Absolute scale comes from the sensor.

Video. No depth, no poses; both estimated. It keeps the one thing the photo tier lacks — continuity — so frames register sequentially into one property-wide cloud and drift correction applies. Keyframes are

sampled at uniform stride *first* and blur-filtered second, so coverage is not biased toward the rooms the operator moved slowly through.

Photo. No depth, no poses, no continuity. Per room: metric depth per image, gravity from the floor plane, scale from camera height, then registration in the three degrees of freedom that survive levelling — yaw from wall-normal histograms, translation by occupancy cross-correlation, then ICP. That order matters: ICP has a small basin of convergence and fails from the identity, and two photographs from opposite corners of a room often share almost no texture but always share the room's shape.

Measured against the operator's tape, which is recorded in whole feet, LiDAR walls come out at a median error of 0.81 m (26%) over 28 walls and photo walls at 2.08 m (75%) over 12. Errors that size are segmentation, merged and short rooms, not sensor noise, which §4 puts under a centimetre.

`capture/DEVICE_MATRIX.md` holds every cell, produced by `scripts/accuracy_table.py` and the gate table rather than written by hand.

3. Drift

The brief makes "poses used as-is" an automatic fail, and it is right to. ARKit's odometry is locally excellent and globally not: on the long walk, 107 m of odometry over 312 s, the pose graph built from its 110 verified revisits starts with a 0.709 m residual.

Correction is a pose graph over keyframes: odometry edges at the reported relative pose, loop-closure edges at the pose ICP measured. **Loop candidates are proposed by geometry and confirmed by ICP, never the reverse.** A candidate must be a genuine revisit — path walked at least $6\times$ the distance closed, and at least 6 m. Without that test a slow walk down a corridor generates a constraint between every pair of keyframes in it, all merely restating odometry with ICP noise added; that mistake produced 114 "closures" and made the map worse. With it, 41 survive on the assignment's scan with ceiling, 19 on its floor-only scan and 110 on the author's long walk.

Rotation and translation residuals are weighted by separate information terms. A pose graph that adds radians to metres is weighting one arbitrarily against the other, and over 100 m a milliradian of yaw costs more than a centimetre of translation. A soft-L1 loss keeps one false closure that survived verification from folding the map; some always survive, because two bathrooms in the same flat look alike to a geometric matcher.

Drift alone does not remove residual yaw, so walls within 6° of the building frame are rotated onto it and their offsets refit from their own points — the plane-anchored half. Only the direction comes from the prior; the position stays measured.

Ablation, 163f18d3ac, the 107 m long walk. All four rows regenerate at this commit from `--no-drift-correction` and `--no-snap-walls`; the last row is the published plan.

variant	rooms	footprint	against tape, 28.75 m ²	loop closures	pose residual	max correction
drift off, snap off	5	20.25 m ²	−30%	0	not applied	—
drift off, snap on	4	16.62 m ²	−42%	0	not applied	—
drift on, snap off	5	25.77 m ²	−10%	110	0.709 → 0.583 m	19.0 cm
drift on, snap on	5	25.27 m²	−12%	110	0.709 → 0.583 m	19.0 cm

Correction is what matters: with snapping on, poses used as-is lose a room and a third of the footprint, and correction recovers both. Snapping does not improve area on this walk. It costs 0.5 m² against the tape with correction, and 3.6 m² and a room without it. It stays on as the plane-anchored half of the correction: the 18 wall runs it rotates sat 1.80° off the building frame on average, which is yaw error if the flat's walls are square, and 0.5 m² is small next to the 8.7 m² between the two walks of this flat. Walls further off than 6° are left as measured, so the long walk's hall is drawn with one wall meeting its neighbours at 78° .

4. Error budget

LiDAR tier, in order of size:

term	magnitude	handling
Residual gravity tilt	not recorded in the plans; 1° tips the far end of a 5 m room by 8.7 cm	Gravity refit to the observed floor plane; a fitted tilt above 6° is not applied and is warned
Yaw drift across rooms	on the long walk 110 closures cut the pose residual from 0.709 to 0.583 m; the 18 wall runs snapped afterwards sat 1.80° off the building frame on average	Pose graph, then walls within 6° snapped onto the frame
Depth noise	8 mm base, +2.5 mm/m ² range term	Inverse-variance weighting; plane σ from the larger of model and residual scatter
Plane offset	0.1–0.4 mm on well-observed walls	Propagated into wall length as $\sqrt{2} \times \sigma$ per corner
Voxel quantisation	20 mm pitch	Positions averaged within voxel, not snapped

Photo tier is dominated by one term that swamps the rest:

term	magnitude
Monocular depth scale	1.57–1.76× over-prediction, measured two ways
Per-frame scale variation	0.71–1.48 across frames
Depth AbsRel vs LiDAR	0.28 mean
Registration	camera separations 0.07–2.52 m, ICP fitness 0.70–0.80 — not a dominant term

The scale term is not a tuning problem. Depth Anything V2 Metric Indoor is trained on normal-field-of-view indoor imagery; the benchmark photographs are 0.5x ultra-wide at 88°. A metric monocular model infers depth from apparent size, which requires an assumed focal length, so an out-of-distribution field of view shifts its metric scale proportionally.

5. Calibration

Intervals are split conformal, fitted per (tier, quantity), with the finite-sample correction — the quantile at $\lceil (n+1)(1-\alpha) \rceil / n$, which is what makes the coverage guarantee hold at small n rather than only asymptotically. Distribution-free, because a wall-length error is a mixture of plane-fit noise, a small scale bias and the occasional gross failure, and no single normal describes that.

Quantities calibrate in the units their error actually scales with: ceiling height in metres, because it is equally hard to measure in a small room or a large one; wall length in percent, because the error grows with the wall.

Where no quantile has been fitted, the interval falls back to the propagated covariance of the fit that produced it, and `Measure.method` says `propagated` or `prior` rather than `conformal`. **That field is load-bearing.** A previous version of `cozmo calibrate` ignored both of its arguments and wrote a fixed table of quantiles labelled as conformal calibration; those flowed into every measurement in every plan and made the one field a reader uses to tell a calibrated interval from a guess into a falsehood. It now fits from residuals or writes nothing and says why.

No quantiles are applied to the published plans. The tape makes fitting possible, and `cozmo calibrate` fits a relative wall-length quantile of 0.60 for LiDAR from 28 residuals, but those are the rows the benchmark scores, so applying it would grade the intervals on their own training data. Every interval in every current plan reports `propagated` or `prior`, truthfully.

6. The fix loop

Full account in `fixloop/`. In brief, because the honest version is the interesting one.

Declared, before the fix: the photo-tier footprint at 142.03 m² against a LiDAR reference of 27.20 m², +422%. Root cause: `intrinsic_from_exif` read only `Image.getexif()`, which returns IFD0 and holds no focal length on an iPhone JPEG; the value is in the sub-IFD at `0x8769`. All 58 photographs carry

`FocalLengthIn35mmFilm = 14`; none was read; every frame ran at the assumed 26 mm prior. `fx` 4125.3 against a correct 2221.3, $1.86\times$ too long. **Predicted: under +50%, and explicitly not a pass.**

Result: +916%. The fix was correct and the number doubled.

The reasoning error: too long a focal length compresses the cloud laterally, which is true, but the room was already too large, so the compression had been partially cancelling a bigger error in depth. $X = (u - cx)Z/fx$, so halving `fx` doubles `X`; the area rose $1.95\times$, which is that almost exactly.

The dominant cause is §4's scale term. Three further changes followed: the scale-correction band widened from [0.75, 1.35] to [0.25, 4.0] — the old band was set while the intrinsics were wrong and was rejecting every correct correction, the prior asking for 0.57 and being refused; room-level scale consensus, since the prior fires on 3 photographs in 20 and scale belongs to the camera rather than to one photograph; and a plausibility guard, because a 113 m² bedroom is not a wide estimate, it is a wrong one, and a large interval does not rescue it.

142.03 → 276.34 → 17.37 m² against 27.20: +422% to −36%. The $\pm 8\%$ gate does not pass, and coverage fell from three wrong rooms to one plausible one. By the brief's own rubric that earns marks for the post-mortem and none for the prediction, which is correct.

Round 2, LiDAR, against the operator's tape. Declared before the fix (88af4e3): footprint −12% on the long walk and −42% on the first walk, every room short, the bedroom 5.28 m² against 9.29 m².

Hypothesis: the strip between furniture and the wall behind it has no floor evidence, so faces there are labelled exterior and rooms end at the wardrobe front; ceiling returns above the strip should count as interior evidence. Predicted: a bedroom of 8.0–9.5 m² and a footprint inside $\pm 5\%$. **Result: every room polygon identical, on both captures.** The "real walls" 0.23–0.27 m beyond the bedroom were the far faces of 230 mm brick partitions, and the ceiling measurement behind the prediction had bled into the neighbouring rooms through a morphological closing. Diagnosing the non-result found a worse defect: the room map had been assigned by matching areas and was wrong on both captures, scoring the long walk's bathroom as the passage and the first walk's bedroom as the hall. Rooms are now named from camera frames. No gate moved.

7. Known failure modes

Twenty are documented with measurements in `known_failure_modes.md`. The four that matter:

The photo tier does not meet its gates. §4 and §6 above. What would fix it, in order: capture at $1\times$ rather than $0.5\times$, which the protocol now requires, though the hall re-shot at $1\times$ still reads 136% too large; fit the focal-to-scale correction against the LiDAR tier, which supplies depth ground truth on the same property for nothing; require two floor-visible photographs per room.

Opening detection barely works at the photo tier. Openings are found by looking for points behind a wall plane that a camera on the near side saw through, and a single photograph's depth map is a 2.5D surface with little behind it. The 58 stills at $0.5\times$ found no opening and the 12 at $1\times$ found one window, so rooms stitch by folder name rather than by doorway. This needs a different detector, not a threshold.

Mirrors, glass and wet-look surfaces get two defences. A geometric mirror test reflects suspect points back across the wall plane and asks whether they land on the room in front of it — a reflection does, a courtyard does not; the sample flat's bathroom walls score 0.21–0.34. And damage requires corroboration from two viewpoints, because a specular highlight is view-dependent and never reprojects to the same patch of surface twice: on the author's marble-and-glass flat that took damage from 21 regions to 1, in a property with none.

Ceiling height is unmeasurable without the upward lap. No downward-facing returns, no height, by any method. The pipeline reports `unmeasured` rather than substituting a default. The company's own `single_room` sample has 61 downward-facing points in the entire scan; the author's first capture had 1.4% of frames aimed up and the second had 24.3%, and the long walk now measures 2.56–2.68 m per room; a window bay that once reported a 1.86 m ceiling, measured from its ledge, now abstains. Reading room levels where each fitted plane crosses the world origin had put that bay at 3.04 m and moved a passage ceiling by 15 cm. Reading the property-wide levels over the floor instead moves the long walk's footprint by 0.76 m², a sensitivity recorded in the failure modes rather than shipped.

8. State of the evidence

The LiDAR tier is the one to run at a walk-in. On the home flat walked with the ceiling lap and a closed loop (163f18d3ac) it returns 5 rooms and 25.27 m² against a taped 28.75 m² (−12%), with per-room ceilings of 2.56–2.68 m, 7 openings and 3 of 4 taped connections. The hall comes closest, 13.18 m² against 14.86 m² (−11%). The bedroom is furthest, 5.28 m² against 9.29 m² (−43%), and a second walk of the same room gives 7.03 m², so the defect is in the reconstruction, not in the tape.

Against the tape the gates read 13 PASS, 19 FAIL, 34 SKIP. Ceiling and opening gates are SKIP because neither was taped. LiDAR intervals cover the tape on none of 36 measurements: they model sensor and drift error, not a merged or a short room. The photo tier fails at +220% and the video tier does not produce a metric plan. Walked alone, the bedroom reads 7.81 m² against 9.29 m²; photographed on the 1× lens, the hall reads 35.12 m² against 14.86 m², down from a rejected 71.8 m² at 0.5×. The assignment's two scans of one flat, which has no tape, give 35.74 m² over 7 rooms and 31.57 m² over 6.

An earlier generator in this repository produced a benchmark over procedurally generated rooms, a head-to-head against an app that was never run, and a fix loop whose before and after were byte-identical. Those artefacts were removed before submission; every number in this report comes from `reports/verified/`.

Some ideas here came from public work on the same brief: a ray-traced test room, a one-command setup script, a relative floor on photo and video intervals, and stitching rooms by folder name when no doorway is matched.